

QNAP QVR

Recording Export

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Contents

History	4
1. FCGI: recording_svc.fastcgi	6
1.1 Recording: Export	7
Export Task Information.....	7
Export Task Request	8
GET	9
Request	9
External URL.....	9
Internal URL	10
Parameter	11
Response	12
Properties.....	13
APIs.....	14
get_shared_folder	14
get_default_dest	15
get_task	16
POST	18
Request	18
External URL.....	18
Internal URL	18
Parameter	19
Request body	19
Properties.....	20
Response	21
Properties.....	22
APIs.....	23
create task	23
PUT	24
Request	24
External URL.....	24
Internal URL	25
Parameter	26
Request body	26
Properties.....	26
Response	27
Properties.....	28

APIs	29
stop	29
clear	30
Pangu	31
Description	31
Format	31
Rule	33
Error Code	34
Internal Errno(error = -1861287672)	35
User Manual	37
Reference	37
1.2 How to add fastcgi	38
Source code	38
Makefile	39
Manifest	40
Shell script	40
Apache	42
Compile	44
SSH	44

History

Revision History	Description	Date	Author
Version 1.0.0 Rev 0.0	1 st version	2023-04-06	Ken Liu
Version 1.1.0 Rev 0.0	1. add "Export Task Information" 2. add "Export Task Request" 3. add GET / POST / PUT APIs 4. add Pangu service	2023-04-12	Ken Liu
Version 1.2.0 Rev 0.0	add retention rule	2023-04-16	Ken Liu
Version 1.3.0 Rev 0.0	fix recording_svc.fastcgi auth	2023-05-03	Ken Liu
Version 1.4.0 Rev 0.0	fix pangu notify content export_information -> iva_info	2023-06-19	Ken Liu
Version 1.5.0 Rev 0.0	1. fix shared folder read / write auth for user 2. add error code 0x91070024, 0x91070025 3. recover task retry max: 2 → 1 4. add more information to task response	2023-07-06	Ken Liu
Version 1.6.0 Rev 0.0	1. support clear task 2. check privilege for stop / clear task 3. refactor state 4. rename error code	2023-07-17	Ken Liu
Version 1.7.0 Rev 0.0	add user manual link	2023-12-12	Ken Liu

Version 1.8.0 Rev 0.0	1. add file_list information 2. check one minute recording data & auto adjust future time	2023-12-29	Ken Liu
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1. FCGI: recording_svc.fastcgi

- External URL must do auth
- Internal URL support auth & no-auth

1.1 Recording: Export

Export Task Information

Name	Type	Default	Description
id	string	empty	task id
account	string	empty	user
created_time	int64_t	0	task created time, UTC timestamp(ms)
execution_start_time	int64_t	0	task start time, UTC timestamp(ms)
execution_end_time	int64_t	0	task end time, UTC timestamp(ms)
state	int	0	0x00: task default(ST_NONE) 0x01: task init(ST_INIT) 0x02: task start(ST_START) 0x04: task stop(ST_STOP) 0x08: task success(ST_SUCC) 0x10: task failure(ST_FAIL) 0x20: task error(ST_ERROR) 0x40: task clear(ST_HIDDEN) 0x7F: all tasks(ST_ALL): 0x01 0x02 0x04 0x08 0x10 0x20 0x40
progress	float	0.0	task progress
file_name	string	empty	task exporting file
file_list	json array	[]	task success file
file_count	int	0	task total file
file_prefix	string	empty	file prefix
total_size	int64_t	0	task total file size
dest	string	empty	task destination
folder	string	empty	task folder
error	int64_t	0	see “Error Code”

log	string	empty	reserve information
retry_count	size_t	0	task retry count

Export Task Request

Name	Type	Default	Description
channel_uuid	string	empty	channel unique id
stream_id	int	0	0: stream 1 1: stream 1 2: stream 2 3: stream 3 255: all streams
rec_type	int	0	0: No_Recording 1: Event Recording 4: Regular Recording 5: All Recording (1 4)
start_time	int64_t	0	export start time, UTC timestamp(ms)
end_time	int64_t	0	export end time, UTC timestamp(ms)
av_only	int	0	0: video + audio 1: video only 2: audio only(not support)

GET

Request

External URL

URI	http://<ip>:<port>/<platform>/recording/v1/func/export? <query_string>
Method	GET
Host	IP address
Port	8080(depend on QTS)
Resource platform	qvrpro, qvrelite etc...
Resource entry	recording
Resource version	v1
Resource object	export
Description	query by act

Internal URL

URI	http://<ip>:<port>/recording/auth/v1/func/export?<query_string> http://<ip>:<port>/recording/v1/func/export?<query_string>
Method	GET
Host	127.0.0.1
Port	11241
Resource entry	recording
Resource auth	auth or empty(no-auth)
Resource version	v1
Resource object	export
Description	query by act

Parameter

Name	Required	Description
@query_string multi-fields: by comma ex: id=aaa,bbb		
qsauth_type	yes	0: sid 1: center token 2: cross token 3: QNE token 4: qsid 5: playback token
qsauth_token	yes	authentication token
order	no	
order_by	no	see “APIs”
limit	no	no more than that many rows will be returned (default: 0)
offset	no	skip that many rows before beginning to return rows (default: 0)
act	yes	see “APIs”
id	no	1. see “Export Task Information” 2. mutli-filed: support
account	no	1. see “Export Task Information” 2. mutli-filed: support 3. auto convert sid → account
state	no	1. see “Export Task Information” 2. mutli-filed: not support

Response

- success

Content-Type	application/json
<pre>{ "success": true, "result": JSON object, array or string, "result_size": >=0, "result_size_unlimited": >=0 }</pre>	

- failure

Content-Type	application/json
<pre>{ "success": false, "error": -1861287672, "error_hex": "0x910F0108", "error_message": "fcgi function exec" (show internal_errno: error = -1861287672) "internal_errno": -1861810167 "internal_errno_hex": "0x91070809" }</pre>	

Properties

Name	Type	Description
success	bool	true or false
result	JSON	JSON object, array or string
result_size	size_t	the number of elements of “result”
result_size_unlimited	size_t	the number of elements of “result” without limit, offset
@see “Error Code”		
error	int	< 0
error_hex	string	error hex string(0x910Fxxxx)
error_message	string	error description
internal_errno	int	< 0(show internal_errno: error = -1861287672)
internal_errno_hex	string	internal error hex string

APIs

get_shared_folder

- available shared folders for user
 - forbidden: read only user
 - forbidden: deny user
- input
 - qsauth_type + qsauth_token
 - act
- output
 - success
 - absolute shared folder path list
 - failure
 - see ["Error Code"](#)
- example
 - `http://ip:port/qvrpro/recording/v1/func/export?qsauth_type=0&qsauth_token=bcw89ix9&act=get_shared_folder`

```
{
  "result": [
    "/share/Container",
    "/share/homes",
    "/share/Multimedia",
    "/share/Public"
  ],
  "result_size": 4,
  "result_size_unlimited": 4,
  "success": true
}
```

get_default_dest

- user used to choose the destination
- input
 - o qsauth_type + qsauth_token
 - o act
- output
 - o success
 - absolute default folder path
 - o failure
 - see ["Error Code"](#)
- example
 - o `http://ip:port/qvrpro/recording/v1/func/export?qsauth_type=0&qsauth_token=bcw89ix9&act=get_default_dest`

```
{
  "result": [
    {
      "account": "admin",
      "dest": "/share/Public/dbg_export",
      "id": "a1fe18b6-4dfb-4cee-b61d-1f1ecfaa5257"
    }
  ],
  "result_size": 1,
  "result_size_unlimited": 167,
  "success": true
}
```

get_task

- show all tasks
 - skip clear task(default)
 - query_string without state
- input
 - qsauth_type + qsauth_token
 - act
 - optional
 - account
 - state
 - multi-state
 - ex: ST_INIT + ST_START + ST_SUCC + ST_STOP
 - 0x0F: 0x01 + 0x02 + 0x04 + 0x08
 - order_by
 - created_time
- output
 - success
 - task information list
 - failure
 - see [“Error Code”](#)

- example

- [http://ip:port/qvrpro/recording/v1/func/export?
qsauth_type=0&qsauth_token=bcw89ix9&act=get_task](http://ip:port/qvrpro/recording/v1/func/export?qsauth_type=0&qsauth_token=bcw89ix9&act=get_task)

```
{
  "result": [
    {
      "account": "admin",
      "created_time": 1681196483796,
      "dest": "/share/Public/dbg_export",
      "error": 0,
      "execution_end_time": 0,
      "execution_start_time": 0,
      "file_count": 2,
      "file_name": "b.mp4",
      "file_list": ["a.mp4", "b.mp4"],
      "file_prefix": "PostmanClient",
      "folder": "",
      "id": "a1fe18b6-4dfb-4cee-b61d-1f1ecfaa5257",
      "log": "",
      "progress": 0.0,
      "state": 3,
      "total_size": 0,
      "retry_count": 0,
      "channel_uuid": "2B3CED70E76A2B3CED70E76A2B3C0000",
      "stream_id": 0,
      "rec_type": 4,
      "start_time": 1687831200000,
      "end_time": 1688501400000,
      "av_only": 0
    }
  ],
  "result_size": 1,
  "result_size_unlimited": 167,
  "success": true
}
```

POST

Request

External URL

URI	http://<ip>:<port>/<platform>/recording/v1/func/export?<query_string>
Method	POST
Host	IP address
Port	8080(depend on QTS)
Resource platform	qvrpro, qvrelite etc...
Resource entry	recording
Resource version	v1
Resource object	export
Description	create export task

Internal URL

URI	http://<ip>:<port>/recording/auth/v1/func/export?<query_string> http://<ip>:<port>/recording/v1/func/export?<query_string>
Method	POST
Host	127.0.0.1
Port	11241
Resource entry	recording
Resource auth	auth or empty(no-auth)
Resource version	v1
Resource object	export
Description	create export task

Parameter

Name	Required	Description
@query_string		
qsauth_type	yes	0: sid 1: center token 2: cross token 3: QNE token 4: qsid 5: playback token
qsauth_token	yes	authentication token

Request body

Content-Type	application/json
<pre>{ "dest": "/share/Public", "file_prefix": "PostmanClient" "channel_uuid": "B9073BCE96E9B9073BCE96E9B9070000", "stream_id": 1, "rec_type": 4, "start_time": 1679468400000, "end_time": 1679472000000, "av_only": 0 }</pre>	

Properties

Name	Type	Description
@see “APIs”		
dest	string	see “Export Task Information”
file_prefix	string	see “Export Task Information”
channel_uuid	string	see “Export Task Request”
stream_id	int	see “Export Task Request”
rec_type	int	see “Export Task Request”
start_time	int64_t	see “Export Task Request”
end_time	int64_t	see “Export Task Request”
av_only	int	see “Export Task Request”

Response

- success

Content-Type	application/json
<pre>{ "success": true "id": "c73ff8b5-a363-4bbc-a94b-74cf28686411" }</pre>	

- failure

Content-Type	application/json
<pre>{ "success": false, "error": -1861287672, "error_hex": "0x910F0108", "error_message": "fcgi function exec" (show internal_errno: error = -1861287672) "internal_errno": -1861810167 "internal_errno_hex": "0x91070809" }</pre>	

Properties

Name	Type	Description
success	bool	true or false
@see "Error Code"		
error	int	< 0
error_hex	string	error hex string(0x910Fxxx)
error_message	string	error description
internal_errno	int	< 0(show internal_errno: error = -1861287672)
internal_errno_hex	string	internal error hex string

APIs

create task

- send request to server, server arranges it into queue
- input
 - o qsauth_type + qsauth_token
 - o see [“Export Task Request”](#)
- output
 - o success
 - task id
 - o failure
 - see [“Error Code”](#)
- example
 - o `http://ip:port/qvrpro/recording/v1/func/export?`
`qsauth_type=0&qsauth_token=bcw89ix9`

```
{
  "dest": "/share/Public/dbg_export",
  "file_prefix": "PostmanClient",
  "channel_uuid": "B9073BCE96E9B9073BCE96E9B9070000",
  "stream_id": 1,
  "rec_type": 4,
  "start_time": 1679468400000,
  "end_time": 1679472000000,
  "av_only": 0
}
```

PUT

Request

External URL

URI	http://<ip>:<port>/<platform>/recording/v1/func/export? <query_string>
Method	PUT
Host	IP address
Port	8080(depend on QTS)
Resource platform	qvrpro, qvrelite etc...
Resource entry	recording
Resource version	v1
Resource object	export
Description	update by act

Internal URL

URI	http://<ip>:<port>/recording/auth/v1/func/export?<query_string> http://<ip>:<port>/recording/v1/func/export?<query_string>
Method	PUT
Host	127.0.0.1
Port	11241
Resource entry	recording
Resource auth	auth or empty(no-auth)
Resource version	v1
Resource object	export
Description	update by act

Parameter

Name	Required	Description
@query_string		
qsauth_type	yes	0: sid 1: center token 2: cross token 3: QNE token 4: qsid 5: playback token
qsauth_token	yes	authentication token

Request body

Content-Type	application/json
<pre>{ "act": "stop", "id": "c73ff8b5-a363-4bbc-a94b-74cf28686411" }</pre>	

Properties

Name	Type	Description
act	string	1. stop / clear 2. see “APIs”
id	string	see “Export Task Information”

Response

- success

Content-Type	application/json
<pre>{ "success": true }</pre>	

- failure

Content-Type	application/json
<pre>{ "success": false, "error": -1861287672, "error_hex": "0x910F0108", "error_message": "fcgi function exec" (show internal_errno: error = -1861287672) "internal_errno": -1861810167 "internal_errno_hex": "0x91070809" }</pre>	

Properties

Name	Type	Description
success	bool	true or false
@see "Error Code"		
error	int	< 0
error_hex	string	error hex string(0x910Fxxx)
error_message	string	error description
internal_errno	int	< 0(show internal_errno: error = -1861287672)
internal_errno_hex	string	internal error hex string

APIs

stop

- stop current task
 - only task owner or administrators
- input
 - qsauth_type + qsauth_token
 - act
 - id
- output
 - success
 - failure
 - see [“Error Code”](#)
- example
 - `http://ip:port/qvrpro/recording/v1/func/export?`
`qsauth_type=0&qsauth_token=bcw89ix9`

```
{
  "act": "stop",
  "id": "a1fe18b6-4dfb-4cee-b61d-1f1ecfaa5257"
}
```

clear

- clear task information
 - o only task owner or administrators
 - o forbidden: ST_INIT or ST_START task
- input
 - o qsauth_type + qsauth_token
 - o act
 - o id
- output
 - o success
 - o failure
 - see ["Error Code"](#)
- example
 - o `http://ip:port/qvrpro/recording/v1/func/export?qsauth_type=0&qsauth_token=bcw89ix9`

```
{
  "act": "clear",
  "id": "a1fe18b6-4dfb-4cee-b61d-1f1ecfaa5257"
}
```

Pangu

Description

- server notify task information to all the clients

Format

- notice_reason
 - o type: 8
 - o action: 8050
- iva_info(export module depends on "iva_info")
 - o task
 - task information list
 - see ["Export Task Infomation"](#) & ["Error code"](#)

```
{
  "iva_info": {
    "task": [
      {
        "account": "admin",
        "created_time": 1681115839485,
        "dest": "/share/Public/dbg_export",
        "error": 0,
        "execution_end_time": 1681115879718,
        "execution_start_time": 1681115839505,
        "file_count": 1,
        "file_name": "PostmanClient_150000_160000.mp4",
        "file_list": ["PostmanClient_150000_160000.mp4"],
        "file_prefix": "PostmanClient",
        "folder": "admin_Ch001_20230410_163719_1681115839485",
        "id": "85b592d6-4e6d-4dd9-855f-a75626defa27",
        "log": "",
        "progress": 100,
        "state": 4,
        "total_size": 1716124677,
        "retry_count": 0,
        "channel_uuid": "2B3CED70E76A2B3CED70E76A2B3C0000",
        "stream_id": 0,
        "rec_type": 4,
        "start_time": 1687831200000,
        "end_time": 1688501400000,
        "av_only": 0
      }
    ]
  },
  "notice_reason": {
    "action": 8050,
    "type": 8
  }
}
```


Retention

Rule

- 1000 count
- 30 days

Error Code

Error	Error(hex)	Description
@Common(0x0000)		
-1861287936	0x910F0000	common error
-1861287935	0x910F0001	split string
@FCGI(0x0100)		
-1861287680	0x910F0100	fcgi common error
-1861287679	0x910F0101	fcgi invalid request uri
-1861287678	0x910F0102	fcgi invalid request method
-1861287677	0x910F0103	fcgi not supported request version"
-1861287676	0x910F0104	fcgi init
-1861287675	0x910F0105	fcgi no stdin data
-1861287674	0x910F0106	fcgi can not find function
-1861287673	0x910F0107	fcgi function not implemented
-1861287672	0x910F0108	fcgi function exec(see "internal errno")
-1861287671	0x910F0109	fcgi not digit
-1861287670	0x910F010A	fcgi invalid auth v3
-1861287669	0x910F010B	fcgi parse json response
-1861287668	0x910F010C	fcgi empty json response
-1861287667	0x910F010D	fcgi not supported json response
-1861287666	0x910F010E	fcgi empty query string
@JSON(0x0200)		
-1861287424	0x910F0200	json common error
-1861287423	0x910F0201	json parse
-1861287422	0x910F0202	json invalid
-1861287421	0x910F0203	json empty data

Internal Errno(error = -1861287672)

Errno	Errno(hex)	Description
@Common(0x0001)		
-1861812223	0x91070001	General error
-1861812222	0x91070002	Empty channel information
-1861812221	0x91070003	Invalid channel id
-1861812220	0x91070004	Invalid channel guid
-1861812219	0x91070005	Invalid stream id
-1861812218	0x91070006	Over max interval(deprecated)
-1861812217	0x91070007	Lower min interval(deprecated)
-1861812216	0x91070008	Can not get file list
-1861812215	0x91070009	No such file or Directory
-1861812214	0x9107000A	Error on file open
-1861812213	0x9107000B	Retry on file open
-1861812212	0x9107000C	Error on file read
-1861812211	0x9107000D	Error on file write
-1861812210	0x9107000E	Error on file release
-1861812209	0x9107000F	Over four hour interval

@Export(0x0020)		
-1861812192	0x91070020	max 20 tasks
-1861812191	0x91070021	create default folder
-1861812190	0x91070022	volume space below 50GB
-1861812189	0x91070023	volume space reaching 95% usage
-1861812188	0x91070024	no available shared folder(deny or empty)
-1861812187	0x91070025	shared folder(read only)
-1861812186	0x91070026	not owner and administrator
-1861812185	0x91070027	forbid clearing ST_INIT or ST_START task
-1861812184	0x91070028	there is a ST_START task running
-1861812183	0x91070029	no read/write permission information from QTS
-1861812182	0x9107002A	QVR is preparing recording data

User Manual

Reference

- <https://ieinet.sharepoint.com/:p:/r/sites/VideoSW1Teams/Shared%20Documents/SW1%20%E6%96%87%E4%BB%B6/API%20%E6%96%87%E4%BB%B6/QNAP%20QVR%20Pro%20%26%20Elite%20MP4%20API%20%E6%93%8D%E4%BD%9C%E7%B0%A1%E4%BB%8B.pptx?d=w35256b3b50b14149b28746516cf38b61&csf=1&web=1&e=oHv0QT>

1.2 How to add fastcgi

- Reference
 - o SurvApp/CmsSvc/IvaSvc
- GitLab
 - o [QVRPRGRD-7322] server export interface(1st commit)
 - <https://sauron.qnap.com/NVR/NVR/SurvApp/-/commit/5cde1aae52fdb16864bf62fa0fca87634235f7df>
- Example
 - o recording_svc.fastcgi
 - o RecordingSvc(ubuntu1804)

Source code

- SurvApp/CmsSvc/RecordingSvc
- main entry
 - o main.cpp

Makefile

- SurvApp/Model/Common/ubuntu_1804/Makefile.CFGSVC

- o CMS_CFGSVC

- RECORDING_REST

```
CMS_CFGSVC:    CFG_PREPARE \
               PRIV_HANDLER \
@@ -67,7 +68,8 @@ CMS_CFGSVC:    CFG_PREPARE \
               TPT_REST \
               CFG_SCRIPT \
               TOOLS_QDB_VALIDATOR \
-             IVA_REST
+             IVA_REST \
+             RECORDING_REST
```

- o CMS_CFGSVC_CLEAN

- RECORDING_REST_CLEAN

```
@@ -83,7 +85,8 @@ CMS_CFGSVC_CLEAN:    CFG_REST_CLEAN \
               MID_REST_CLEAN \
               IDX_REST_CLEAN \
               TOOLS_QDB_VALIDATOR_CLEAN \
-             IVA_REST_CLEAN
+             IVA_REST_CLEAN \
+             RECORDING_REST_CLEAN
```

- o .PHONY

- RECORDING_REST

```
+ RECORDING_REST:
+     @echo "making $@....."
+     cd $(CMS_ROOT)/RecordingSvc; \
+     sudo make CROSS_COMPILE=$(CROSS_COMPILE) NVR_LINKFS=$(NVR_LINKFS) CFG_ROOT=$(CFG_ROOT) ARCH_DEF=$(ARCH_DEF)
+     NASWARE_PATH=$(NASWARE_PATH) HTTP_ROOT=$(HTTP_ROOT) FASTCGI_LIB_VERSION=$(FASTCGI_LIB_VERSION)
+     SYS_LIB_PATH=$(SYS_LIB_PATH) APACHE_SRC=$(APACHE_SRC) TARGET_PREFIX=$(TARGET_PREFIX)
+     BASE_UBUNTU_VER=$(BASE_UBUNTU_VER) MANIFEST_CONF_DEF=$(MANIFEST_CONF_DEF) MANIFEST_CONF=$(MANIFEST_CONF) ARCH=$(ARCH);
```

- RECORDING_REST_CLEAN

```
+ RECORDING_REST_CLEAN:
+     @echo "make clean $@....."
+     cd $(CMS_ROOT)/RecordingSvc; \
+     sudo make clean;
```

Manifest

- SurvApp/Model/Common/ubuntu_1804/Makefile.config.def
 - o CFG_ENABLED_RECORDING_SVC ?= yes
- SurvApp/Model/Common/ubuntu_2204/Makefile.config.def
 - o CFG_ENABLED_RECORDING_SVC ?= yes
- SurvApp/Model/Common/Makefile.config.def
 - o CFG_ENABLED_RECORDING_SVC ?= no
 - o not support ubuntu1604

All the “Makefile.config.def” must set default, otherwise qpkg will build error

```
@@ -38,6 +38,7 @@ THIRDP_ENABLED_DOOR_CTL ?= no
# CFG_ENABLED_SREST fastcgi for STRDB (sw1)
# CFG_ENABLED_IDXSVC fastcgi for recording db (sw1)
# CFG_ENABLED_IVASVC fastcgi for IVA (sw1)
+ # CFG_ENABLED_RECORDING_SVC fastcgi for recording function (sw1)
CFG_BUILD_MSGHDLER ?=yes
CFG_ENABLED_MSGHDLER ?=yes
CFG_BUILD_DBGTRM ?=yes
@@ -46,6 +47,7 @@ CFG_ENABLED_SREST ?= yes
CFG_ENABLED_IDXSVC ?= no
CFG_ENABLED_UUIDGENGUID ?= no
CFG_ENABLED_IVASVC ?= yes
+ CFG_ENABLED_RECORDING_SVC ?= yes
```

Shell script

- SurvApp/RootFS/init.d_509/cfg_service.sh
 - o exec it by manifest “CFG_ENABLED_RECORDING_SVC”
 - register_fcgi()
 - /sbin/daemon_mgr.qvrpro recording_svc.fastcgi file_start
"/Qtmp/.recording_svc_fault"

```
+ if com_is_enabled "CFG_ENABLED_RECORDING_SVC"; then
+   /sbin/daemon_mgr.qvrpro recording_svc.fastcgi file_start "/Qtmp/.recording_svc_fault"
+ else
+   log_msg "CFG_ENABLED_RECORDING_SVC=[no]"
+ fi
```

- unregister_fcgi()
 - /sbin/daemon_mgr.qvrpro recording_svc.fastcgi file_stop
"/Qtmp/.recording_svc_fault"


```
+   if com_is_enabled "CFG_ENABLED_RECORDING_SVC"; then
+       /sbin/daemon_mgr.qvrpro recording_svc.fastcgi file_stop "/Qtmp/.recording_svc_fault"
+   else
+       log_msg "CFG_ENABLED_RECORDING_SVC=[no]"
+   fi
+
```

- kill recording_svc.fastcgi

```
+   if com_is_enabled "CFG_ENABLED_RECORDING_SVC"; then
+       /bin/kill -9 ` /bin/pidof recording_svc.fastcgi ` >& /dev/null
+   fi
+
```

Apache

- SurvApp/SysSvc/HTTP/apache_php_conf/apache_u1804.conf

- port

- 11241(discuss port number with your team leader)

```
+ Listen 127.0.0.1:11241
```

- entry(add rules by yourself)

- VirtualHost

- ServerName RecordingService
 - ScriptAlias /recording/ /usr/local/cmsbin/
 - ErrorLog "|/usr/bin/rotatelog /hd/logs/error_recording_log.%w 86400"
 - #CustomLog "|/usr/bin/rotatelog /hd/logs/access_recording_log.%w 86400" combined
 - CustomLog /dev/null common

- Directory

- AllowOverride None
 - Options +ExecCGI -MultiViews +SymLinksIfOwnerMatch
 - Require all granted
 - RewriteEngine on
 - RewriteBase /recording/
 - RewriteCond %{REQUEST_FILENAME} !-f
 - RewriteRule .* /recording/recording_svc.fastcgi?

- auth(if needed)
 - Order deny,allow
 - Deny from all
 - Allow from 127.0.0.1
 - SetEnvIf Authorization "(.+)" HTTP_AUTHORIZATION=\$1

```
+ <VirtualHost _default_:11241>
+   ServerName RecordingService
+   ScriptAlias /recording/ /usr/local/cmsbin/
+
+ <Directory "/usr/local/cmsbin">
+   Order deny,allow
+   Deny from all
+   Allow from 127.0.0.1
+   #AuthBasicAuthoritative Off
+
+   AllowOverride None
+   Options +ExecCGI -MultiViews +SymLinksIfOwnerMatch
+   Require all granted
+
+   SetEnvIf Authorization "(.+)" HTTP_AUTHORIZATION=$1
+
+   RewriteEngine on
+   RewriteBase /recording/
+   RewriteCond %{REQUEST_FILENAME} !-f
+   RewriteRule .* /recording/recording_svc.fastcgi? [L]
+ </Directory>
+
+   ErrorLog "||/usr/bin/rotatelog /hd/logs/error_recording_log.%w 86400"
+   #CustomLog "||/usr/bin/rotatelog /hd/logs/access_recording_log.%w 86400" combined
+   CustomLog /dev/null common
+ </VirtualHost>
```

- Apache config(QTS)
 - SurvApp/Others/QPKG/qpkgDev/apps/<platform>/src-shared/etc/apache-qvrpro.conf
 - platform
 - QVR Pro

```
+ ProxyPass /qvrpro/recording http://127.0.0.1:11241/recording/auth retry=0
```

Compile

- cd SurvApp/Model/<platform>
 - o platform
 - QVR Pro: SS-ALL
 - QVR Elite: QELT
- make BASE_UBUNTU_VER=18.04 RECORDING_REST
- make BASE_UBUNTU_VER=18.04 RECORDING_REST_CLEAN

SSH

- fastcgi path
 - o /usr/local/cmsbin/recording_svc.fastcgi
- fastcgi reload
 - o /etc/init.d/cfg_service.sh stop
 - o /etc/init.d/cfg_service.sh start